Automated Stacking of Irregularly Shaped Stones via GPU-Accelerated Pose Optimization and Monte Carlo Tree Search

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***Abstract*—Stacking irregularly shaped objects into a freestanding structure of a prescribed shape is challenging because each placement changes both occupied space and contact force distribution throughout the structure, potentially affecting structural stability. Even a currently stable placement can restrict subsequent placements, limiting how closely the completed structure approximates the target shape. Existing approaches based on masonry heuristics evaluate stability through simulation, but do not refine poses according to contact force redistribution throughout the structure or explicitly compare subsequent placement sequences. We propose a bilevel pose optimizer that uses differentiable support functions to compute pose gradients accounting for target geometry and force and torque equilibrium across the entire structure. Furthermore, we develop a Monte Carlo Tree Search (MCTS) framework with hierarchical node expansion that samples and refines candidate placements, then validates them through simulation to compare alternative placement sequences. To address the computational burden of MCTS for this fundamentally combinatorial problem, we parallelize contact force optimization within each structure using bodywise ADMM and batch pose optimization and simulation across independent candidates on the GPU. We present preliminary examples of equilibrium optimization across the entire structure and placement-sequence search, alongside a physical stacking demonstration.**

I. Introduction

Stacking is a fundamental problem with applications such as bin packing, industrial warehouse storage, and stone-wall construction. In robotic bin packing, container walls can provide lateral support [21, 9]. Without such support, freestanding stacking of n irregular objects must approximate a prescribed target shape $T \subset \mathbb{R}^3$ while maintaining force and torque equilibrium under gravity through inter-object and ground contacts. Satisfying these requirements is challenging because pose changes of irregular objects alter both occupied space and contact locations and normals, coupling geometric fit with force and torque equilibrium across the entire structure. For a completed structure, this goal can be expressed as minimizing unfilled target volume subject to physical feasibility:

$$\min_{q_1,\cdots,q_n \in \mathrm{SE}(3)} \mathbb{V}\left(T \setminus \bigcup_{i=1}^{n} O_i(q_i)\right) \quad \text{s.t.} \quad c_{\text{phys}}(O_1(q_1),\cdots,O_n(q_n)) = 0, \tag{1}$$

where $O_i(q_i) \subset \mathbb{R}^3$ is the region occupied by object i at pose $q_i \in \mathrm{SE}(3)$, $\mathbb{V}(\cdot)$ denotes volume, and $c_{\text{phys}} = 0$ represents stability and non-interpenetration constraints. Constructing such a structure also requires selecting objects, determining which objects support one another, and deciding in what order to place them. Each placement changes the remaining inventory, available space, and support for subsequent objects, so a currently stable placement may limit the final target coverage. These interdependent discrete choices make sequence planning combinatorial, while evaluating candidate placements requires iterative numerical evaluation of contact interactions [7], making the search computationally expensive.



Fig. 1: Preliminary stacking results. Left: simulated structures for different target shapes. Right: an outdoor stacking experiment using a CPU-only implementation due to platform constraints. The limited computational budget precluded the disturbance-based robustness reward (Sec. V-D).

However, existing automated approaches to stacking irregular objects make limited use of structural stability information to refine poses and give limited consideration to subsequent placement sequences. Johns et al. [12, 13] use masonry heuristics to guide stone arrangement in freestanding walls, together with geometric alignment of stone surfaces and simulation-based stability checks. Liu and Napp [16] select placements through heuristic filtering and simulation-based assessments of structural strength. Although these approaches evaluate structural stability, they do not use force and torque equilibrium across the entire structure to guide continuous pose refinement. Their consideration of future placements relies on masonry heuristics rather than explicit comparison of alternative placement sequences. We address these limitations through pose optimization incorporating structural stability and GPU-accelerated Monte Carlo Tree Search (MCTS) for sequence planning (Figure 2).

Unlike prior stacking methods, we formulate placement refinement as a constrained bilevel problem, following for-

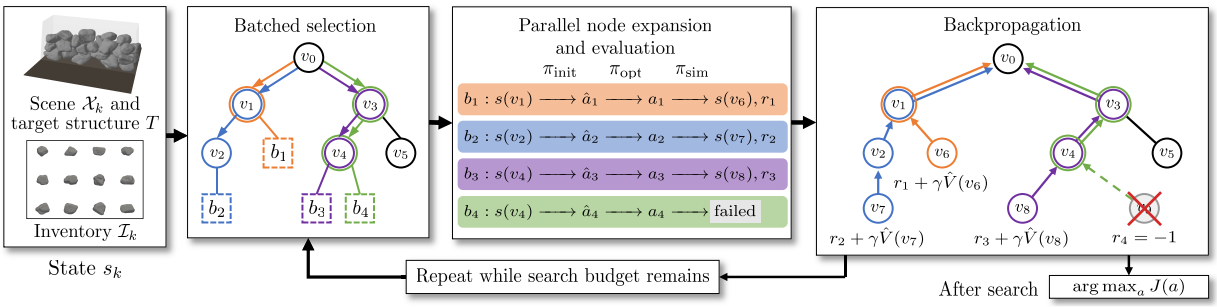


Fig. 2: Batched MCTS sequence planning. Colored requests are evaluated in parallel. Returns combine rewards and discounted leaf estimates. The dashed branch backpropagates its failure penalty once ($r_4 = -1$, $\hat{V}(v_9) = 0$) and is never selected again. After search, $J(a)$ ranks eligible root actions. See Sec. V-A for $\hat{V}(v)$ and $J(a)$.

mulations used in contact-aware manipulation planning and grasp synthesis [5, 15, 22]. The lower-level problem optimizes contact forces across all stones, including the candidate, to evaluate force and torque balance throughout the structure, with shared contact forces coupling contacting bodies. The upper-level problem refines the placement pose using gradients of this optimized physical objective together with geometric objectives, enabled by differentiable support functions (DSFs) and the associated collision detection [14, 2].

In addition, rather than relying solely on masonry heuristics, we use MCTS to plan stacking sequences, explicitly accounting for how each placement affects subsequent placements. To adapt MCTS to stacking, the proposed hierarchical node expansion strategy progressively refines and validates candidate placements through initial pose sampling, pose optimization, and simulation-based feasibility checking. Domain-specific rewards further guide the search toward target shape coverage and structural stability.

Repeated pose optimization and simulation make MCTS computationally demanding. Recent robotics research exploits GPU parallelism to accelerate optimization [20, 5] and physical simulation [11]. Following this direction, we introduce bodywise contact force copies and an alternating direction method of multipliers (ADMM) to decompose the coupled force optimization into parallel bodywise updates, with consensus enforcing consistency between the copies. Combined with GPU-accelerated subADMM simulation, this enables batched pose optimization and simulation across independent candidates to accelerate MCTS. Our main contributions are as follows:

- A differentiable bilevel pose optimization formulation guided by the target shape T and force and torque equilibrium across the entire structure.
- An MCTS framework for stacking with hierarchical node expansion and domain-specific rewards guiding lookahead toward the target coverage and structural stability.
- GPU-accelerated MCTS evaluation through bodywise ADMM for parallel contact force optimization within structures and batched pose optimization and simulation across candidates.

II. Related Works

A. Robotic Packing and Stone Stacking

Packing and stacking combine geometric criteria with stability assessment. For container packing, Wang and Hauser [21] evaluate pile equilibrium via linear programming. For free-standing walls, Johns et al. [12, 13] combine masonry rules, candidate contact force optimization, geometric alignment, and simulation. Liu et al. [17] geometrically filter simulation-settled poses, with Liu and Napp [16] adding shake tests for structural strength. Menezes et al. [18] learn placement values reflecting future rewards with fixed object sequences and orientations.

B. Contact-Aware Pose Optimization

Bilevel manipulation and grasp planners refine configurations using optimized contact forces. Lee et al. [15] use contact factors and the envelope theorem under a predefined task skeleton. GraspQP [22] differentiates the Karush–Kuhn–Tucker (KKT) conditions of a force-closure quadratic program. BODex [5] combines bilevel grasp optimization with GPU-batched force solves, using finite differences or detached contact points for contact refinement. We differentiate DSF contact geometry at regular solutions to account for force and torque equilibrium across the entire structure.

C. Simulation-Based Sequential Manipulation Planning

PMBS [8] batches GPU physics simulations during MCTS expansion and rollout for object retrieval from clutter. We adopt batched expansion, adding parallel pose refinement with contact force optimization for the entire structure before simulation.

III. Preliminaries and Overview

A. Shape Representation

We represent the target shape T and each object O_i as unions of smooth convex bodies modeled by differentiable support functions (DSFs) [14]. For a compact convex body $C \subset \mathbb{R}^3$, its support function is $h_C(x) = \max_{p \in C} x^\top p$ for $x \neq 0$. At pose $q = (R, t) \in \mathrm{SE}(3)$, we write

$$C(q) = \{Rp + t \mid p \in C\}, \qquad O_i(q_i) = \bigcup_j C_{ij}(q_i),$$

where C_{ij} are the convex components in object i's local frame. Similarly, $T = \bigcup_j T_j$, where T_j are fixed convex components in the world frame. DSFs support differentiable signed-distance and contact-geometry computations between posed convex bodies [2], enabling gradient-based pose optimization.

B. Markov Decision Process Formulation

Given an inventory of n objects, we model stacking as a finite-horizon sequential decision process in which one object is placed at each step. Let $\mathcal{P}_k \subseteq \{1, \dots, n\}$ be the indices of the objects already placed, and let $i_0 \notin \{1, \dots, n\}$ index the fixed ground. The current scene is $\mathcal{X}_k = \{(i, q_i) \mid i \in \mathcal{P}_k\}$, and the remaining inventory is $\mathcal{I}_k = \{1, \dots, n\} \setminus \mathcal{P}_k$. The state, action, and deterministic transition function are

$$s_k = (\mathcal{X}_k, \mathcal{I}_k, T), \quad a_k = (i_{k+1}, q_{k+1}), \quad s_{k+1} = F(s_k, a_k),$$

where $i_{k+1} \in \mathcal{I}_k$. The transition inserts the proposed pose into the scene, simulates the resulting structure, and returns the settled scene with $\mathcal{I}_{k+1} = \mathcal{I}_k \setminus \{i_{k+1}\}$. The reward $r(s_k, a_k)$ scores target-contained volume and physical quality; feasibility checks separately reject invalid transitions (Sec. V). The sequence planner therefore operates on the MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, F, r, \gamma)$.

C. Monte Carlo Tree Search

Monte Carlo Tree Search (MCTS) incrementally constructs a search tree $\mathcal{T} = (\mathcal{V}, \mathcal{E})$ over the MDP. Each node $v \in \mathcal{V}$ represents a state $s(v)$, and each edge $(v, a) \in \mathcal{E}$ represents an action and its transition to a child node. Let $N(v)$ denote the number of completed evaluations and $Q(v)$ the empirical mean of discounted returns backpropagated to that node.

Each iteration selects a path by balancing value and exploration, expands and evaluates an action, and backpropagates its discounted return to update Q and N.

In our planner (Fig. 2), expansion uses three modules: π_{init} proposes an object and initial pose $\hat{a}_k = (i_{k+1}, \hat{q}_{k+1})$, π_{opt} refines the pose, and π_{sim} evaluates the resulting structure to determine feasibility, the settled successor, and reward. The resulting flow is

$$s_k \xrightarrow{\pi_{\text{init}}} \hat{a}_k \xrightarrow{\pi_{\text{opt}}} a_k \xrightarrow{\pi_{\text{sim}}} s_{k+1}, \; r(s_k, a_k)$$

and the details are introduced in the later sections.

IV. Pose Optimization

The module π_{opt} refines $\hat{q}_{k+1}$ from π_{init} toward a stable, physically feasible placement q_{k+1}.

A. Single Object Pose Optimization Problem

Given s_k, we initialize q at $\hat{q}_{k+1}$ and solve the bilevel problem

$$\begin{aligned} \min_{q} \quad & \left[f_{\text{geo}}(q) + \min_{\lambda} f_{\text{phys}}(q, \lambda) \right] \\ \text{s.t.} \quad & g_{\text{cone},j}(q, \lambda_j) \le 0, \quad j = 1, \cdots, n_c \end{aligned} \tag{2}$$

where $q \in \text{SE}(3)$ is the candidate object's pose and $\lambda = (\lambda_1, \cdots, \lambda_{n_c}) \in \mathbb{R}^{3n_c}$ collects all contact forces in the resulting structure, including inter-object and ground contacts. This

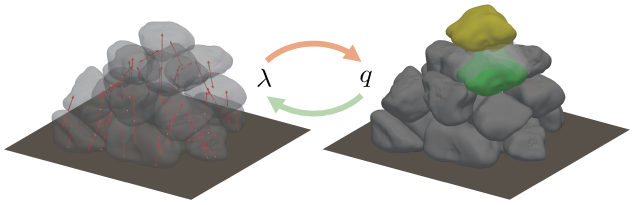


Fig. 3: Bilevel pose optimization. Left: optimizing contact forces λ under friction constraints minimizes structural equilibrium residuals at fixed q. Right: refining $q \in \text{SE}(3)$ minimizes the geometric and optimized physical objectives.

formulation follows prior contact-aware optimization methods [15, 5].

The geometric objective f_{geo} is designed as follows:

$$\begin{aligned} f_{\text{geo}}(q) = w_{\text{h}} f_{\text{h}}(q) + w_{\text{in}} f_{\text{in}}(q \mid T, O_{i_{k+1}}) \\ + w_{\text{pen}} f_{\text{pen}}(q \mid \mathcal{X}_k, O_{i_{k+1}}) \end{aligned}$$

with positive weights. For $q = (R, t)$, $f_{\text{h}}(q) = t_z$ is the world-frame height of the candidate's pose origin. Writing $O = O_{i_{k+1}}$, we penalize proximity to existing objects and the fixed ground as

$$f_{\text{pen}}(q) = \sum_{j \in \mathcal{P}_k \cup \{i_0\}} \left[\epsilon_{\text{margin}} - \text{d}(O(q), O_j(q_j)) \right]_+^2,$$

where d is signed distance (positive for separation), $[x]_+ = \max(x, 0)$, and $\epsilon_{\text{margin}} \ge 0$ is the clearance margin. With $h_{O(q)}(u) = \max_j h_{C_{i_{k+1},j}(q)}(u)$, we define

$$\begin{aligned} m_{\text{in}}(q) &= \max_j \min_{u \in \mathbb{S}^2} \left[h_{T_j}(u) - h_{O(q)}(u) \right], \\ f_{\text{in}}(q) &= \left[-m_{\text{in}}(q) \right]_+^2. \end{aligned}$$

This term penalizes violations of containment of the entire object within a single convex target component, a conservative condition for containment in T. Noting that the term is differentiable almost everywhere in q, we compute pose derivatives at regular inner solutions by implicit differentiation. Nonsmooth transitions have not caused practical difficulties in our implementation.

Let $\mathcal{P}_{k+1} := \mathcal{P}_k \cup \{i_{k+1}\}$ and let $I_l \subseteq \{1, \dots, n_c\}$ index all contacts involving object l, including ground contacts. The physical objective excludes the ground's equilibrium residual:

$$\begin{aligned} f_{\text{phys}}(q, \lambda) = w_{\text{eq}} \sum_{l \in \mathcal{P}_{k+1}} \| J_{c, I_l}^{\top}(q) \lambda_{I_l} + \lambda_{\text{ext},l} \|_W^2 \\ + w_{\text{cp}} \sum_{j \in I_{i_{k+1}}} \text{d}_j(q)^2 \|\lambda_j\|^2 + w_{\text{reg}} \sum_{j=1}^{n_c} \|\lambda_j\|^2 \end{aligned} \tag{3}$$

The three terms respectively penalize static equilibrium residuals, violations of contact complementarity between the candidate and the current scene, and large contact forces. Figure 4 illustrates the effect of evaluating equilibrium across the entire structure on candidate placement. The weight matrix $W \succ 0$ balances force and moment scales. Here, $\text{d}_j(q)$ is

the signed distance between the bodies of contact candidate j; J_{c,I_l}, $\lambda_{\text{ext},l}$, and λ_{I_l} are object O_l's contact Jacobian, external wrench, and contact force vector, respectively; and $w_{\text{eq}}, w_{\text{cp}}, w_{\text{reg}} > 0$ are objective weights.

Finally, the smoothed friction cone constraint at contact j is $g_{\text{cone},j}(q, \lambda_j) \leq 0$, where

$$g_{\text{cone},j}(q, \lambda_j) = \sqrt{\|J_{c,j}^t \lambda_j\|^2 + \epsilon} - \mu_j J_{c,j}^n \lambda_j \tag{4}$$

Here, $\epsilon > 0$ is a smoothing parameter, μ_j is the friction coefficient, and $J_{c,j}^t$ and $J_{c,j}^n$ are the tangential and normal components of the Jacobian of the contact constraint.

B. Upper-Level Optimization on SE(3)

Let $\lambda^*(q)$ be the solution of the lower-level problem and define $f_{\text{phys}}(q) := f_{\text{phys}}(q, \lambda^*(q))$. The envelope theorem [4] gives

$$\frac{df_{\text{phys}}(q)}{dq} = \frac{\partial f_{\text{phys}}(q, \lambda^*)}{\partial q} + \sum_{j=1}^{n_c} \nu_j^* \frac{\partial g_{\text{cone},j}(q, \lambda_j^*)}{\partial q}$$

where ν_j^* is the optimal dual variable for the j-th friction cone constraint. Using this gradient, we optimize the placement pose q on SE(3) using a Riemannian trust-region method [1] with a BFGS Hessian approximation [10]. A homotopy schedule adjusts the geometric and physical weights between stages, holding them fixed within each stage and initializing from the preceding pose. It progressively reduces w_{reg} to limit regularization-induced equilibrium residuals. Candidate evaluation uses the final pose and forces.

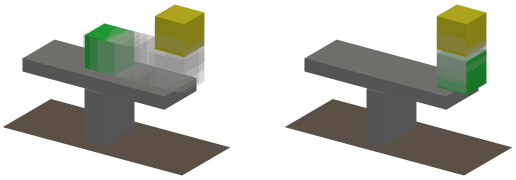
C. Lower-Level Problem Optimization with ADMM

For fixed q, the lower-level problem is convex in λ. To apply bodywise ADMM [3], we introduce two local copies $\lambda_{j,1}$ and $\lambda_{j,2}$ per contact, one for each contacting body, including the ground. Collecting these copies into $\tilde{\lambda}$ makes each body's equilibrium residual depend only on its local variables. Consensus and friction constraints give the reformulation

$$\begin{aligned} \min_{\tilde{\lambda}, z} \quad & \tilde{f}_{\text{phys}}(q, \tilde{\lambda}) + \sum_{\substack{j=1,\dots,n_c \\ s=1,2}} \delta_{K_j}(z_{j,s}) \\ \text{s.t.} \quad & \lambda_{j,1} = \lambda_{j,2} = z_{j,1} = z_{j,2}, \qquad j = 1, \dots, n_c. \end{aligned}$$

where $\tilde{f}_{\text{phys}}$ is the physical objective in local variables, $z_{j,s}$ are auxiliary variables, and δ_{K_j} is the indicator of $K_j := \{\lambda \in \mathbb{R}^3 \mid g_{\text{cone},j}(q, \lambda) \leq 0\}$. ADMM solves the body subproblems independently. An equivalent four-dimensional second-order cone (SOC) lifting enables closed-form cone projections for the friction updates, while consensus updates reduce to averaging.

The pose optimization module is implemented using CUDA to leverage GPU parallelization for efficient computation of the gradients and the ADMM updates.



(a) Structure-wide (b) Candidate-only

Fig. 4: Seesaw example with initial (yellow), optimized (green), and intermediate (gray) poses. Evaluating equilibrium across the entire structure (a) moves the candidate toward the support. Candidate-only equilibrium (b) leaves it near the unsupported end.

V. Sequence Planning via Parallel MCTS

Building on Sec. III, we combine batched MCTS [8] with parallel pose optimization, sampling, simulation, and domain-specific rewards (Fig. 2). In receding-horizon planning, we search a limited horizon, execute one selected placement, and replan from the observed scene.

A. Batched Search Procedure

Batched Selection. Colored paths in the "Batched selection" panel (Fig. 2) terminate at expansion requests b_j. Over selectable child actions of v_p, we define

$$P(v_p, a) = \text{softmax}_a(w_O \rho_O^{\text{pre}}(a) - w_E e(a)),$$

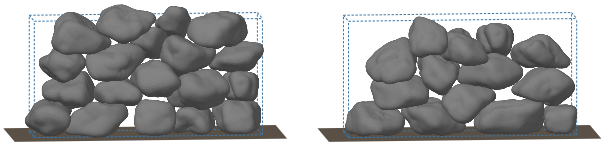
where $\rho_O^{\text{pre}}(a)$ is the pre-simulation object-volume fraction inside the target; $e(a)$ and the weights follow Sec. V-D. Writing $N_v = N(v)$ and similarly for $Q, \hat{N}$, we select evaluated children using PUCT [19]:

$$\text{PUCT}(v) = \frac{N_v Q_v}{N_v + \hat{N}_v} + cP(v_p, a)\frac{\sqrt{N_{v_p} + \hat{N}_{v_p}}}{1 + N_v + \hat{N}_v}, \tag{5}$$

where $c > 0$ controls exploration, $\hat{N}$ counts temporary reservations, and the first term is zero when its denominator is zero. We use progressive widening [6] during tree search to expand the set of candidate placements as visits accumulate. Selection follows evaluated, feasible children using PUCT until a node is chosen for expansion. Reservations prevent duplicate batch evaluations without increasing the widening allowance.

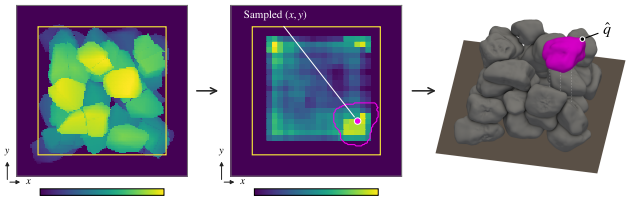
Parallel Expansion and Evaluation. Each colored row in Fig. 2 evaluates one request through $\pi_{\text{init}} \to \pi_{\text{opt}} \to \pi_{\text{sim}}$; rows run in parallel using the pose optimizer of Sec. IV.

Leaf Value Estimation and Backpropagation. For non-terminal, feasible leaves, we replace rollouts with $\hat{V}(v) = \hat{r} \sum_{j=0}^{d_v - 1} \gamma^j$, where d_v is capped by remaining search depth and inventory. The fixed hyperparameter $\hat{r}$ sets a heuristic per-step reward for unexplored continuations, influencing exploration through backpropagated returns alongside PUCT. Failed or terminal nodes have $\hat{V} = 0$. Following the arrows in the "Backpropagation" panel (Fig. 2), we propagate $r_v + \gamma \hat{V}(v)$, with incoming reward r_v, toward the root, adding discounted ancestor rewards, updating Q, N, and clearing reservations.



(a) MCTS (b) Greedy

Fig. 5: Wall construction using the same inventory of over 40 stones. MCTS places 19 stones (44.0% target filling); greedy stops at 14 (36.1%), finding no further feasible placement. Blue dashed outlines show the target.



(a) Scene height (b) Proposal score (c) Sampled initial pose

Fig. 6: Initial pose sampling for a fixed object and orientation: (a) scene height, (b) proposal score and sampled planar location, and (c) initial pose $\hat{q}$ above the scene.

Action Selection and Tree Reuse. As shown at the bottom right of Fig. 2, we select the feasible root action maximizing

$$J(a) = (1-\beta)Q(v_a) + \beta G(a), \qquad \beta \in [0,1].$$

Here $G(a)$ is the maximum discounted observed return over the deepest evaluated feasible paths below a, including its reward but excluding bootstrap values; root-only paths are eligible. This term offsets exploration failures that depress Q despite promising paths. If no root action is feasible, no placement is returned.

After execution, the observed successor becomes the root. We retain descendants and statistics only if the observed state and planning context match the subtree, discarding it otherwise.

Figure 5 compares the resulting planner with greedy placement.

B. Initial Pose Sampling

The sampler π_{init} selects objects with volume bias and uses finite orientation sets, seeding empty scenes near feasible target corners. For object i, orientation R, and feasible planar location $\mathbf{x}$, the score (Fig. 6) is

$$\sigma(\mathbf{x} \mid i, R) = w_{\text{s}}\sigma_{\text{s}} + w_{\text{f}}\sigma_{\text{f}} + w_{\text{b}}\sigma_{\text{b}} - w_{\text{h}}\sigma_{\text{h}} - w_{\text{d}}\sigma_{\text{d}},$$

with positive weights. The terms measure predicted bottom support fraction, unfilled layer area beneath the footprint, feasible-boundary proximity, normalized mean contact height, and a bounded footprint-gap penalty to the preceding object (zero for the first placement). Following Menezes et al. [18], floor and boundary preferences encourage filling empty regions and building inward from the boundary.

We combine elite selection with sampling without replacement weighted by $\exp(\sigma/\tau)$, $\tau > 0$. We initialize height so the candidate's lowest point lies above the base plane and the bounding-box tops of objects with overlapping padded XY bounds, with added clearance (Fig. 6(c)). Geometric filters constrain planar displacement and placement height after refinement and settling.

C. Simulation Module

At every search depth, π_{sim} first settles the candidate for a fixed time duration with existing objects fixed, then resets velocities and simulates the entire structure. Settling requires a minimum duration and sustained low mass-normalized kinetic energy; excessive accumulated motion, invalid contacts, or timeout reject the placement.

The settled scene defines $s' = F(s,a)$. From rest in s', independent disturbance trials apply a force of magnitude $\alpha m_{i_{k+1}} g$ at the candidate's center of mass for Δt_f, in n_d evenly spaced horizontal directions ($\alpha > 0$, gravitational acceleration g). With all objects dynamic, each trial must settle again without exceeding prescribed displacement limits relative to s'; trials do not alter s'. GPU-accelerated subADMM [11] evaluates independent candidates in parallel.

D. Reward Design

For the new object $i = i_{k+1}$ at its settled successor pose q_i' (Sec. V-C), define

$$\rho_T = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(T)}, \quad \rho_O = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(O_i)}.$$

These voxel-estimated volume ratios reward target filling and containment. For feasible transitions, with fixed normalization scales absorbed into positive weights,

$$\begin{aligned} r(s,a) = {} & w_T\rho_T + w_O[\rho_O - \tfrac{1}{2}]_+ - w_E e(a) \\ & + w_H\big(\phi_H(s) - \phi_H(s')\big) - w_D D, \end{aligned}$$

where $\phi_H(s) = \text{Var}_\Omega(H_s)$ is the area-weighted variance of scene height over the target footprint Ω, with empty cells assigned zero height. Coverage lost through earlier objects' motion is not subtracted. Failed transitions receive $r = -1$.

For pose-optimization residual forces $\mathbf{f}_l$ and torques $\boldsymbol{\tau}_l$ across all placed objects,

$$e(a) = \max_l \left(\frac{\|\mathbf{f}_l\|}{m_l g} + \frac{\|\boldsymbol{\tau}_l\|}{m_l g \mathbb{V}(O_l)^{1/3}} \right).$$

Candidates with $e(a) \geq \epsilon_{\text{eq}}$ are rejected before simulation. Disturbance sensitivity measures motion from settled poses q_l' during force trials:

$$D = \min\left\{ D_{\max}, \max_{j,l,t} d_{\text{SE(3)}}\big(q_l^{(j)}(t), q_l'\big) \right\},$$

Here $q_l^{(j)}(t)$ is object l's pose in trial j at time t. The maximum spans all trials, objects, and simulation times including recovery. The distance $d_{\text{SE(3)}}$ sums translation distance and angular distance weighted by $\lambda_\theta > 0$.

Acknowledgments

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